

The Cramér Conjecture in O'Neil Harmonic-Mean Form

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Attribution

Building on Cramér (1936), Shanks (1964), and Granville (1995), this work proposes a sharpened form of Cramér's strong conjecture with constant equal to the current empirical record $R_{\max} = 0.9206386$, expressed via the harmonic-mean construction $H(n) = m - 1/m$ derived from the $2p$ factoring-free framework. The conjecture asserts that the current record is asymptotically optimal — a stronger claim than Cramér's original ($C = 1$) and contrary to Granville's heuristic prediction ($\limsup R = 2e^{-\gamma} \approx 1.1229$).

Statement

For every even integer $n > 10^6$, let $m = n/2$ and define the harmonic mean of $(m+1)$ and $(m-1)$:

$$H(n) = 2 / (1/(m+1) + 1/(m-1)) = m - 1/m$$

Define the proximity to the nearest prime ($\ln$ = natural log):

$$d(n) = \min_{\{p \text{ prime}\}} |H(n) - p|$$

Then:

$$d(n) < 0.4603193 \cdot (\ln n)^2 \quad (1)$$

Corollary (Prime Gaps)

If p_k and p_{k+1} are consecutive primes with $p_k < m < p_{k+1}$, then (1) implies:

$$p_{k+1} - p_k < 0.9206386 \cdot (\ln p_k)^2 \quad (2)$$

Equivalently, the Cramér-Shanks-Granville ratio $R(p) = g(p)/(\ln p)^2$ satisfies $R(p) < 0.9206386$ for all primes $p > 10^6$.

Empirical Support

The conjecture is consistent with all known prime gaps up to 4×10^{18} (Oliveira e Silva, Herzog & Pardi 2014, *Math. Comp.*) and extended to $\approx 1.84 \times 10^{19}$ (Prime Gap Search project, 2024). The current record $R = 0.9206386$ was discovered by Bertil Nyman in 1999 at $p = 1,693,182,318,746,371$ with $\text{gap} = 1132$, and has stood for 27 years despite extensive search.

Notes

- $H(n)$ avoids integer divisibility by construction (it is $m - 1/m$, never an integer for $m > 1$), so (1) is naturally framed as a claim about the 'stubborn FP' residual class after Stage-1 sieving in the $2p$ factoring-free framework.
- The conjecture is strictly stronger than Cramér's original ($C = 1$) and strictly weaker than the 'no upper bound' view. It pins the bound to the observed empirical record rather than to a heuristic constant.
- **Falsifiable** by a single counterexample: any prime p with gap g such that $g \geq 0.9206386 \cdot (\ln p)^2$ would disprove (2) and hence (1).
- The lower bound $n > 10^6$ excludes small-scale anomalies where R can fluctuate above 0.92 due to sparse-prime regions. Above this bound, all empirical data confirm $R < 0.9206386$.